

Ashish Kumar Singh

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WORK EXPERIENCE

Bharat Electronics Limited (BEL)

August 2025 – Present

Software Development Engineer

New Delhi, IN

- Owned and delivered **10+ backend modules** and **20+ REST APIs** for **Umbrella NMS**, covering monitoring, configuration, alarms, reporting, RBAC, audit workflows, and a concurrent multi-protocol poller; supported **211 device types**, **2200+ operational interfaces**, and **5M+ daily telemetry records**.
- Implemented **Redis caching** for current-device telemetry using cache-aside access, explicit invalidation, and polling-aware TTLs, achieving a **85% cache hit rate**, reducing database load from **850 to 180 QPS (~79%)** and cutting **P95 API latency from 240 ms to 52 ms (~78%)**.
- Implemented an event-driven report-processing pipeline for **ORION** using **Spring Kafka**, with **reportId**-keyed ordering, independent consumer groups, transactional outbox, retry/DLT handling, and idempotent at-least-once processing; **recovered an 80K-message backlog in under 5 minutes**.
- Optimized **Hibernate/JPA and SQL** across **18+ high-volume workflows**, eliminating N+1 query paths and reducing average query latency by **~32%**; improved representative execution from **125 ms to 82 ms** through indexing, batching, fetch-strategy tuning, and **EXPLAIN ANALYZE**.
- Containerized and deployed **independently deployable Spring Boot services** on **Amazon EKS** via ECR/Helm; provisioned AWS infrastructure with Terraform across EKS, RDS PostgreSQL, MSK, ElastiCache, S3, IAM, VPC, and ALB, supporting **4× peak workload** while reducing **idle compute by ~35%** through HPA/KEDA autoscaling and Karpenter-based node capacity management.
- Built **CI/CD and GitOps delivery** for microservices using **GitHub Actions**, **GitHub OIDC**, and **Argo CD**, automating unit/integration tests, security scans, immutable SHA-tagged image publishing to ECR, and promotion across development, staging, and production with approval-gated releases and rollback to known-good images.

Defence Research and Development Organisation (DRDO)

June 2024 – August 2024

Software Developer Intern

New Delhi, IN

- Engineered an **eBPF-based TCP packet filtering system** at the NIC level using C, libbpf, and Linux APIs, improving filtering efficiency by **~37%**.
- Optimized kernel-level packet processing and benchmarked multiple traffic patterns, reducing network latency by **25%** while improving throughput and stability under load.

TECHNICAL SKILLS

Languages: Java, C++, C, Python, SQL, JavaScript

Backend & Distributed Systems: Spring Boot, Spring Security, Hibernate/JPA, REST APIs, Apache Kafka, Spring Kafka, Redis, PostgreSQL, MySQL, Microservices, Event-Driven Architecture

Cloud & DevOps: AWS (EKS, ECR, RDS, S3, MSK, ElastiCache, IAM, VPC, ALB), Docker, Kubernetes, Terraform, Helm, GitHub Actions, Argo CD

Testing & Security: JUnit 5, Mockito, MockMvc, Testcontainers, JaCoCo, OIDC, JWT, RBAC, TLS/mTLS, AWS Secrets Manager

EDUCATION

National Institute of Technology, Jalandhar

Jalandhar, IN

Bachelor of Technology, Computer Science and Engineering

2021 - 2025

PROJECTS

AI Inference Gateway | Java, Spring Boot, Docker, Nginx, Ollama, llama.cpp, REST APIs



- Built a self-hosted **AI inference gateway** exposing unified Spring Boot REST APIs across Ollama/llama.cpp backends, with Dockerized deployment and Nginx-based routing for multi-model inference.

RoleMate | Java, Spring Boot, WebSocket, WebRTC, PostgreSQL, React, Docker



- Built a real-time interview platform with **scored matchmaking**, **concurrent queues**, **WebSockets**, **WebRTC**, **PostgreSQL**, and **React**; containerized the stack with Docker and added **55+ tests** for backend reliability.